

**Normal distribution** A symmetric continuous probability distribution that adheres to the familiar bell-shaped curve; has many convenient properties and is useful for drawing statistical inferences.

**Out-of-sample** All new data set that is completely different from the one over which one optimizes a strategy or builds a model.

**Overconfidence** Subjective belief that one's own judgment is better than it really is; the majority believe they are superior to the average; this can cause investors to underreact to new information.

**Overfitting bias** When a statistical model is excessively complex and the model describes random error or noise more than an underlying relationship; characterized by poor predictive performance.

**Overreaction** To react disproportionately to new information; leads to past winners being overpriced and past losers being underpriced.

**Overspecification** See *Overfitting bias*.

**Passive investment management** A predetermined strategy that attempts to mirror a benchmark index; also known as a buy-and-hold approach.

**Private equity** Capital for investment directly into private companies not quoted on a public exchange.

**Prospect theory** Explains why individuals make decisions that deviate from rational decision making by observing how they perceive expected outcomes: people value gains differently than they do losses and prefer to base decisions on perceived gains rather than perceived losses.

**Quartiles** Splitting data into four equal parts.

**Quintiles** Splitting data into five equal parts.

**Random walk** In finance, the theory stating stock prices are independent and unpredictable; it is consistent with the efficient market hypothesis.